



Q1. I would like to know how to read the program already loaded into the microcontroller. What are the software/hardware to be used for the same? Before loading a new program code, what is there inside the memory of a fresh microcontroller?

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A1. Reading the microcontroller is normally done through both software and hardware. There are thousands of different types of microcontrollers available in the market but the procedure to read the data in the memory is almost the same. Typical microcontrollers commonly available in the market are shown in Fig. 1. For example, if you want to read PIC microcontroller, you can use MPLAB IDE and PIC Start Plus programmer board. After launching the MPLAB IDE, you can read the MCU memory contents by selecting the 'Programmer/Read' or 'View/Program Memory' options from the menu bars. The microcontroller having blank or empty space in its memory as read by MPLAB IDE will be as shown in Fig. 2. A microcontroller that has been already loaded with program code will be as shown in Fig. 3.

Normally, a fresh microcontroller chip will have blank flash memory but with some data like signature byte, device ID, calibration byte, etc in separate memory locations. For instance, all microcontrollers manufactured by Atmel Corporation have a three-byte signature code in each device for identification. The code resides in a separate memory address space. The code is written in the memory of the chip during the manufacturing process and cannot be changed. Every chip having the same device number has the same signature byte. For example, all the chips with ATmega16 device number will have the same signature byte. The details regarding the signature byte and its memory

location may be given in the datasheet of respective devices. There are some software which can read them directly.

The signature bytes can be read even if the device is locked and also in

both serial and parallel modes. By reading this code, it is possible to get details about the chip such as manufacturer, device number, etc. For example, the ATmega16 microcontroller has the signature bytes at the memory locations as given in the table.

You can verify these values using a suitable programmer board and software. If you have a microcontroller programmer that supports TopView Programmer software from Front Line Electronics, you can check the signature byte and also read the memory straight away. To read the signature byte, insert the ATmega16 into the programmer, select 'Device' option from the menu bar and click 'Read Signature Bytes' option as shown in Fig. 4.

Signature Bytes of ATmega16	
Signature byte	Detail
\$000: \$1E	Indicates manufactured by Atmel
\$001: \$94	Indicates 16kB flash memory
\$002: \$03	Indicates ATmega16 device when \$001 is \$94

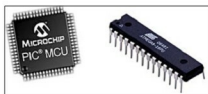


Fig. 1: Typical microcontroller chips

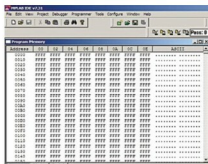


Fig. 2: Blank memory

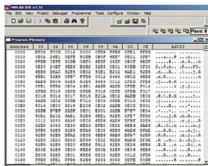


Fig. 3: Program memory filled with data

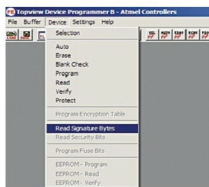


Fig. 4: Reading signature bytes

Q2. Can you suggest me a suitable PCB layout designing software?
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A2. The kind of software tool for designing the PCB layout depends on your requirement and applications. If you are a hobbyist or a student planning to use it for small projects, then you do not need a high-end EDA tool. There are a lot of open source as well as proprietary software available on the Internet. The open source EDA tools such as gEDA and KiCad are quite popular now. If you are interested in proprietary software, you may go for Altium Designer (former Protel) and PCAD by Altium or EAGLE by CadSoft Computer or Proteus by Labcenter Electronics. You may refer to comparison of various EDA tools from the link below. http://en.wikipedia.org/wiki/Comparison_of_EDA_software

The answer to this related query on PCB designing software had been already covered in Qs&As section of EFY February 2013 and May 2013 issues. So, much detail on this topic regarding the use of any of this software to make PCB layout is not discussed here.

Answers compiled by EFY Lab. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender